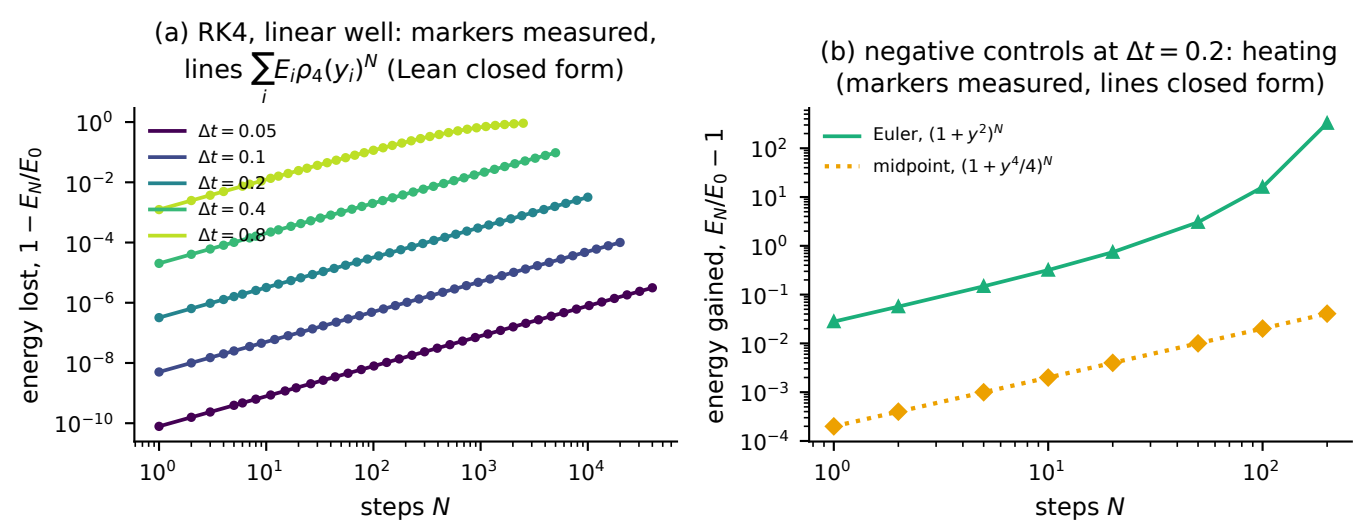
(a) RK4, linear well: markers measured,
lines $\sum_i E_i \rho_4(y_i)^N$ (Lean closed form)
energy lost, $1 - E_N/E_0$
steps N
$\Delta t = 0.05$
$\Delta t = 0.1$
$\Delta t = 0.2$
$\Delta t = 0.4$
$\Delta t = 0.8$
(b) negative controls at $\Delta t = 0.2$: heating
(markers measured, lines closed form)
energy gained, $E_N/E_0 - 1$
steps N
Euler, $(1 + y^2)^N$
midpoint, $(1 + y^4/4)^N$